

Causal Inference HW2

This assignment examines the problem of endogeneity in the estimation of the return to education and evaluates the validity of instrumental variables using the **mroz** dataset. All analysis should be conducted in R using the **wooldridge** and **AER** packages. Present your code, output, and concise interpretations for each question.

1. Data Preparation (Load the dataset **mroz** from the **wooldridge** package)

- Keep only observations where the woman participates in the labor force ($inlf = 1$).
- Display the first two rows of the resulting dataset and report the total number of remaining observations.

2. Estimate the following model

$$lwage \sim educ + exper + expersq + kidslt6 + kidsge6 + husage + huswage + city$$

Report the estimated coefficient on *educ* and interpret it in terms of percentage change in *lwage* per additional *educ*.

3. Suppose *educ* is endogenous. Using the following instruments

$$motheduc, fatheduc, huseduc$$

Estimate the model using two-stage least squares (2SLS) via **ivreg()** function from the **AER** package.

- Report the IV estimate for *educ*. Compare the IV estimate to the OLS estimate from Question 2. Which estimate is larger?
- What does this imply about the possible direction of OLS bias (upward or downward)?
- Report the first-stage F statistics for the instruments. Based on the rule of thumb (< 10 indicates weak instruments), are the instruments weak?

4. From the IV regression output, locate the **Wu-Hausman** test result.

- State the null and alternative hypotheses clearly.
- Report the test statistics and p-value. Based on the p-value, what do you conclude? Is *educ* endogenous or exogenous in this dataset?

5. Recall that a model is overidentified when the number of IVs exceeds the number of endogenous regressors.

- Is the IV model from Question 3 overidentified? Explain briefly.
- Locate the **Sargan** test. State the null and alternative hypotheses for this test. Based on the test's p-value, what do you conclude about the instrument validity?